

ConnectHub - Product Requirements Document (PRD)

Version 1.0 | 5 Sep 2026 | Soumil Gurjar | Django + Tailwind + PWA

ConnectHub Product Requirements Document (PRD)

****Version:**** 1.0 ****Date:**** 5 Sep 2026 ****Owner:**** Soumil Gurjar (India) ****Stack:**** Django 4.x + DRF + SQLite / PostgreSQL + Tailwind

1. Vision

****One-liner:**** Instagram + Twitter ka simple, developer-friendly MVP for portfolio. Friendly, youthful, creator-focused.

****Goal:**** Working MVP in 46 weeks, deployable to Railway/Render, mobile PWA ready.

****Success Metrics:**** SignupPost <2 min, Feed load <1s, Like/Comment <300ms, DM polling 2s, 0 critical 500s.

2. Scope

MVP (Must Have)

- Auth: signup/login (email+password, no verification), logout
- Profile: unique username (slugified, 30), bio 150, avatar, followers/following counts, display_name, interests, onboarding 3-steps
- Post: content 500 + 14 images (5MB each), soft delete, location/tagged
- Feed: followed + own, reverse chronological, 20/page, infinite scroll, stories (24h), right-rail suggested
- Follow: unidirectional, self-block, counts
- Like: toggle, unique (user,post)
- Comment: 200 chars, 1-level nested, edit/delete within 5 min
- DM: 1-to-1, polling 2s, WebRTC call/VC (signaling via polling), story views tracking
- Explore: search (people/tags/captions), trending hashtags, grid masonry
- Settings: 6 sections (Account/Privacy/Notifications/Security/Appearance/About) permanent save
- API: DRF for all above + search, conversations, messages
- Landing (optional), Reels (vertical 9:16), Notifications (bell)

Nice-to-Have (if time)

- Trending, suggested, dark mode, admin ban/delete

Out of Scope (v2+)

Video posts, stories reels, group chat, hashtags trending algo, mobile native, monetization

3. Personas

- ****Gen-Z Creator (1824):**** posts photos, follows creators, DMs friends
- ****Portfolio Reviewer:**** checks code, design system, deploy

4. Tech Stack

Backend Django 4.x/6.x, DRF, Pillow, python-dotenv, corsheaders, whitenoise, gunicorn, dj-database-url, psycopg2-binary, django-storages+boto3 (S3/R2), Channels optional. DB SQLite local, Postgres prod. Media local dev, S3 prod. Frontend HTML5 + Tailwind CDN, vanilla JS. Deploy Railway/Render (Dockerfile, Procfile). PWA manifest + sw.js.

5. Data Models

****Profile**** user O2O cascade, username unique slug 30, bio 150, avatar, followers/following counts, interests, display_name, onboarding_done, created_at
****Post**** author FK cascade, content 500, location 100, tagged 200, is_deleted bool, created_at
****PostImage**** post FK cascade, image, order
****Follow**** follower/following FK cascade, Unique(follower,following), Check follower!=following
****Like**** user/post FK cascade, Unique(user,post)
****Comment**** user/post FK cascade, parent_comment FK self null, content 200, created_at, updated_at
****Message**** sender/receiver FK cascade, content 1000 (blank allowed), image nullable, is_read, created_at; get_conversation(Q)
****Story**** user FK, image, created_at, expires_at (now+24h), is_expired()
****StoryView**** story FK, viewer FK, unique(story,viewer)
****CallSignal**** caller/receiver FK, type offer/answer/ice/hangup, data, is_video, is_read

6. Key Flows

Signup Onboarding 1 (find people) 2 (interests) 3 (photo/bio) Feed
Login Feed (stories + posts) Create Detail Like/Comment (5min) Follow DM (polling + call/VC) Explore/Search Profile (followers/following, tagged, suggested, story views) Settings Reels (vertical, any reel, share to WA/FB/Snap) Notifications Logout

7. API (base /api/, Session auth, PAGE_SIZE 20)

POST /api/signup/, POST /api/login/, GET /api/profile/me/, GET|PUT /api/profile/<username>/, GET|PUT /api/post/s/, GET /api/posts/<id>/, DELETE (soft), GET /api/feed/, POST /api/posts/<id>/like/, GET|POST /api/posts/<id>/comments/, POST /api/follow/<username>/, GET /api/search/?q=, GET /api/conversations/, GET|POST /api/messages/<username>/?after=<id>, POST /api/messages/<username>/signal/, POST /api/messages/<username>/ack/

8. UI Design System

Brand friendly/youthful: Primary #7C3AED, Secondary #06B6D4, Neutrals #111827/#6B7280/#F9FAFB, Success #10B981
Warning #F59E0B Error #EF4444. Fonts Sora (head) + Inter (body) + Grand Hotel (logo). Spacing 4/8/12/16/24/32
/48. Components: btn primary/secondary/ghost, input, card, avatar gradient ring, story ring, post card. Dark v
ia class, Tailwind dark:. Layout: sidebar 220 + feed 630 + right 320; mobile topbar + bottom nav; stories hori
zontal, reels vertical snap.

9. Non-Functional

Perf <1s feed, <300ms like, Security HSTS/SSL/secure cookies/CSRF, S3 for media, PWA installable, a11y labels,
5MB image limit.

10. Timeline (6 weeks)

W1 Setup+Auth+Profile, W2 Posts, W3 Feed+Follow, W4 Like+Comment, W5 DM+Call/VC+Stories, W6 Polish+Deploy+Docs

11. Risks

Media ephemeral without S3, polling vs WebSocket, spam without rate-limit, 404 for expired stories handled as
friendly page.

12. Future

S3 done, PWA done, Docker done; next: WebSocket Channels, push, moderation, app (Expo).

2026 ConnectHub Soumil Gurjar